

ECE 3441 Spring 2025

Quiz 2

Version B

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 60 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

SOLUTIONS

First Name Last Name, Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		30
2		20
3		15
4		25
5		20

Total: _____/(100 possible points + 10 bonus points)

Question #1:

Given the following equation:

$$F(A, B, C, D) = \sum m(1, 3, 7, 15) + \sum d(0, 5, 6, 13, 14)$$

Find the minimum POS by only using a K-Map:

		00	01	11	10
00		X	1	1	0
01		0	X	1	X
11		D	X	1	X
10		0	0	0	0

$$F = (D)(A' + B)$$

Question #2:

Show whether the following is true or false. Hint: truth table; K-Maps might be harder.

$$\overbrace{A'BC + AB'C' + ABC' + ABC}^{F_1} = \overbrace{(A+B+C)(A'+B+C')(A+B'+C)(A+B+C')}^{F_2}$$

~~Handwritten scribble~~

$$A'BC + \cancel{AB'C'} + \cancel{ABC'} + AB'C'$$

$$A'BC + AB + AB'C'$$

$$\cancel{AB+BC} + \cancel{AB'C}$$

$$\cancel{B(A+C)} + \cancel{AB'C'}$$

$$AB + BC + AB'C'$$

A	B	C	F ₁	F ₂
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1	1	
1	0	0	1	
1	0	1		0
1	1	0	1	
1	1	1	1	

$$\therefore F_1 = F_2$$

TRUE

Question #3:

Each of three coins has two sides, heads and tails. Represent the heads or tails status of each coin by a logical variable (A for the first coin, B for the second coin, and C for the third) where the logical variable is 1 for heads and 0 for tails. Write a logic function $F(A, B, C)$ which is 1 only if exactly one of the coins is tails after a toss of the coins.

Express F :

a) As a minterm expansion:

$$F = 011 + 101 + 110 = \sum m(3, 5, 6)$$

$$= ABC' + AB'C + A'BC$$

b) As a maxterm expansion:

$$F = \prod M(0, 1, 2, 4, 7)$$

$$= (A+B+C)(A+B+C')(A+B'+C)(A'+B+C)(A'+B'+C')$$

Question #4:

Using K-Maps, given $F(A, B, C) = A(B + C)$: $\Rightarrow AB + AC'$

a) Express F' as a minterm expansion (use m -notation and $\sum m(\dots)$ notation)

Instead of 1 put down 0

AB \ C	0	1
00	0	1
01	0	1
11	0	0
10	0	1

$$F' = 000 + 001 + 010 + 011 + 101$$

$$= \sum m(0, 1, 2, 3, 5)$$

Ignore the boxes
NOT looking for
minimum SOP

b) Express F' as a maxterm expression (use M -notation and $\prod M(\dots)$ notation)

$$F' = \prod M(4, 6, 7)$$

Question #5:

Find the minimum SOP of the following expression using only K-Maps.

$$F(A, B, C, D) = A'B'D' + BD + ABD' + AB'$$

AB \ CD	00	01	11	10
00	0	0	0	1
01	0	1	1	0
11	1	1	1	1
10	1	1	1	1

$$F = B'D' + BD + A$$